CODEHIVE

JAVA PROGRAMMING: FLOW CONTROL & LOGIC

▸ Introduction to Conditions

Conditions allow a program to make decisions. Java chooses what to execute based on whether a **boolean expression** is **true** or **false**.

Real-world Logic: "If it is 10 PM, then sleep; else, keep studying."

▸ The if Statement

```
if (condition) {
    // code runs if condition is true
}

// Example
if (20 > 18) {
    System.out.println("Access Granted");
}
```

▸ Caution: Braceless IF

If you omit braces, only the **one single line** immediately following the `if` belongs to the condition.

```
if (20 > 18)
    System.out.println("Standard usage"); // Only this is conditional
```

Recommendation: Always use curly braces `{ }` to prevent bugs.

▶ The else & else if Statements

Use `else` for a default path and `else if` to test multiple specific conditions.

```
int time = 22;

if (time < 10) {
    System.out.println("Good morning");
} else if (time < 18) {
    System.out.println("Good day");
} else {
    System.out.println("Good evening");
}
```

▶ Advanced: Ternary Operator

A short-hand replacement for simple if...else blocks. Syntax: `(cond) ? trueVal : falseVal`

```
String result = (time < 18) ? "Day" : "Night";
```

▶ Logical Compounders

- **&& (AND):** Both conditions must be true.
- **|| (OR):** At least one condition must be true.
- **! (NOT):** Reverse the boolean value.

```
if (age >= 18 && isCitizen) {
    System.out.println("Eligible to vote");
}
```

▸ Decision Making Summary

- **if:** Primary entry point for any logic.
- **else if:** Subsequent chainable checks.
- **else:** The final "catch-all" block.
- **Nested IFs:** Allows for layered "if inside an if" decisions.